Hidden multivariate patterns to locate cognitive events on a trial-by-trial basis

Gabriel Weindel

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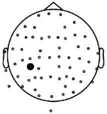
What are event related potentials?



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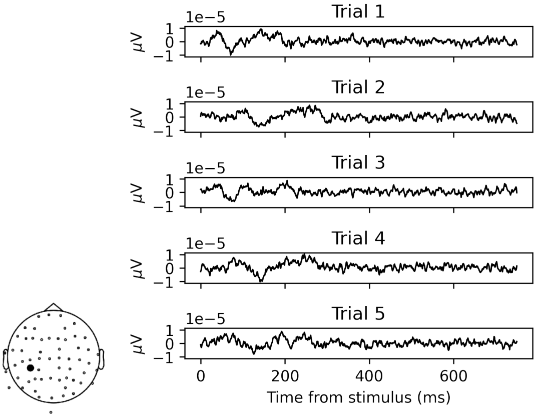

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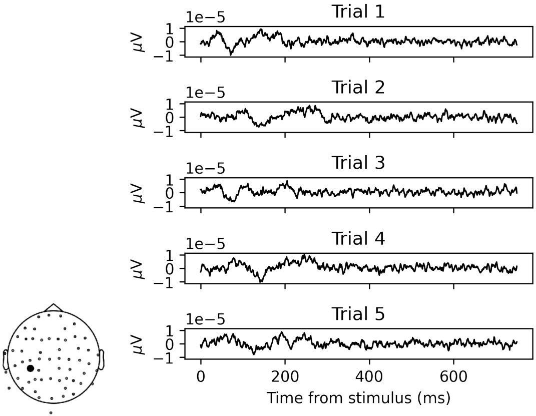

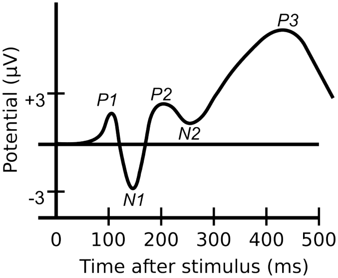

Figure: Wikimedia

It is about time!

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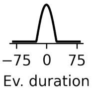
- We don't know the single-trial ERP generators



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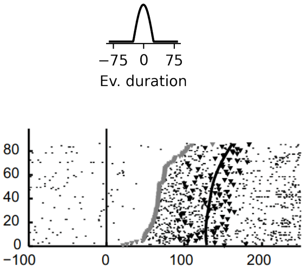

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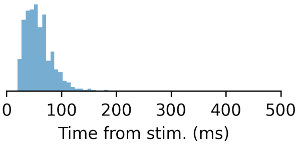

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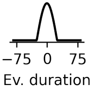

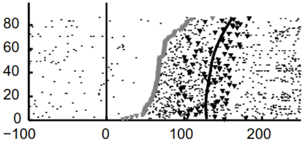

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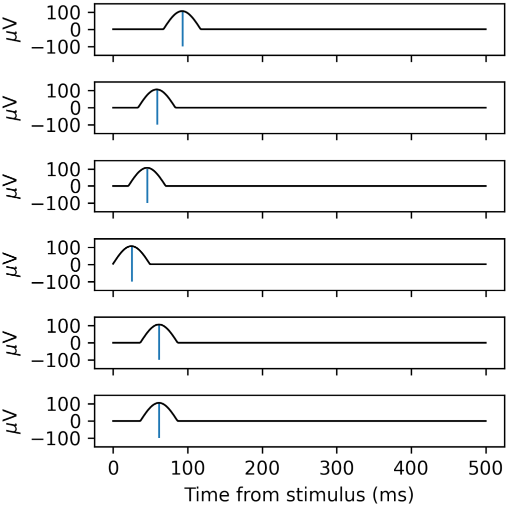

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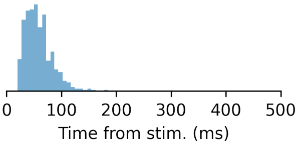

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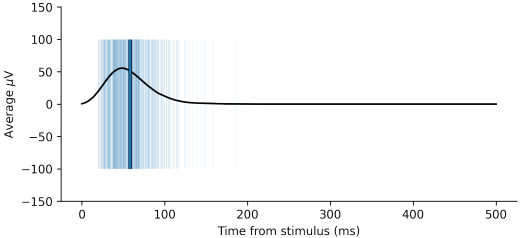

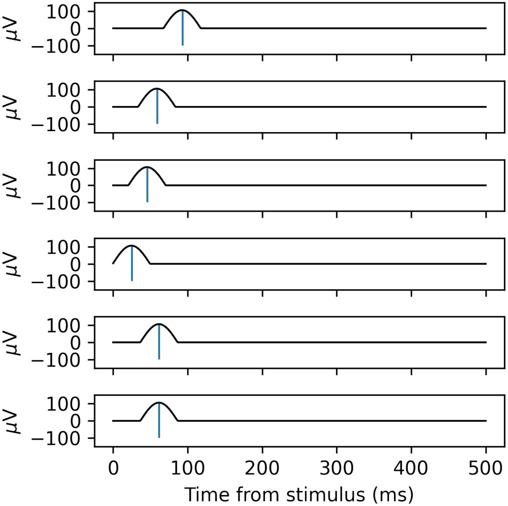

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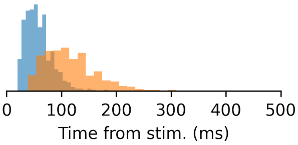

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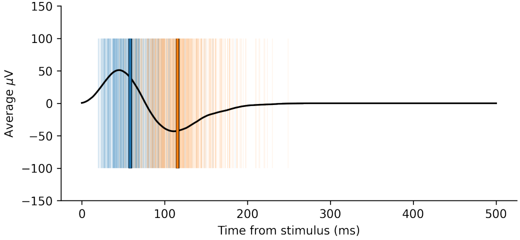

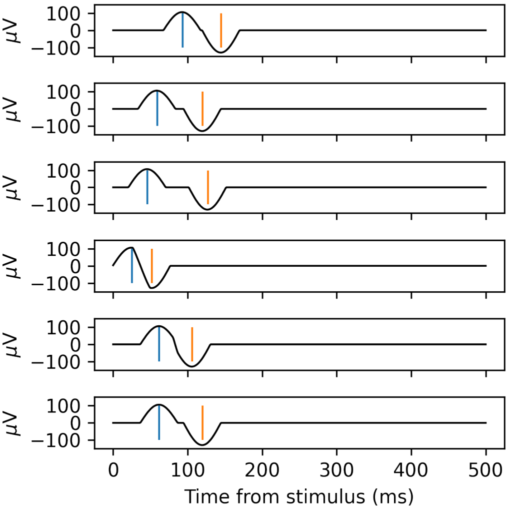

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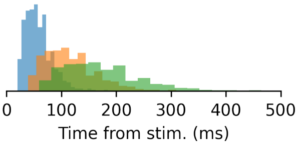

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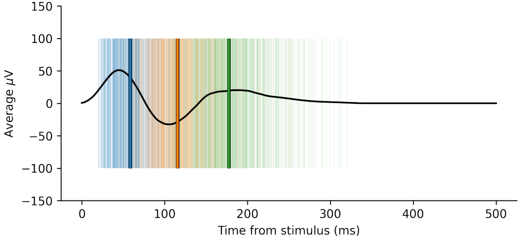

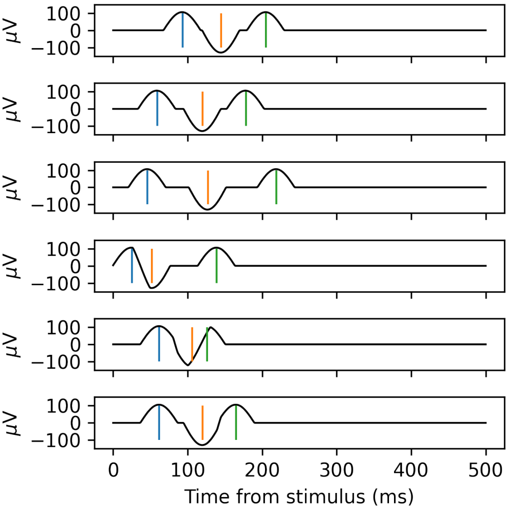

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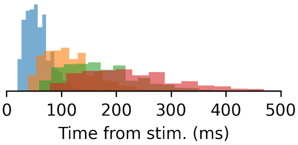

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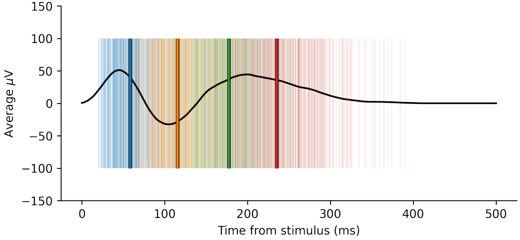
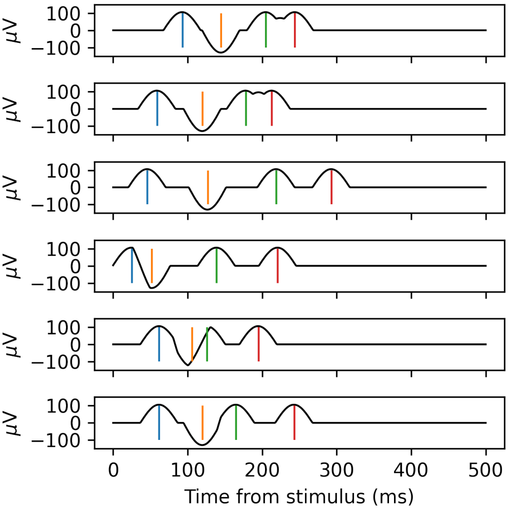
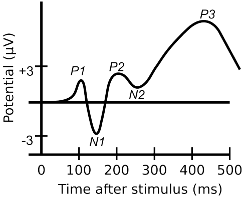




Inferring from average ERPs

Source: Wikimedia commons



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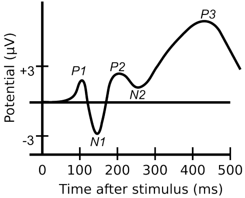
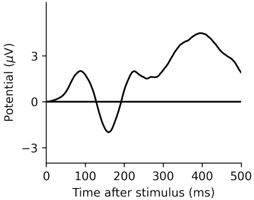


Source: Simulation



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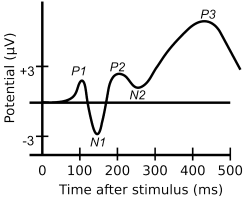

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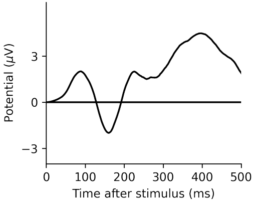

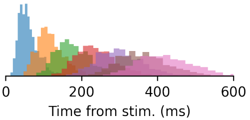

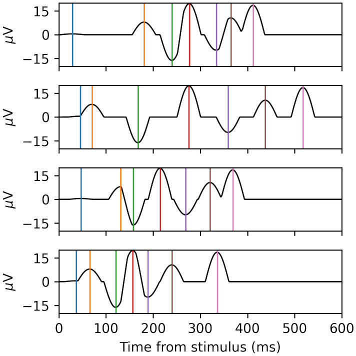

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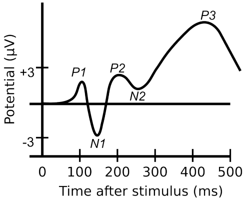

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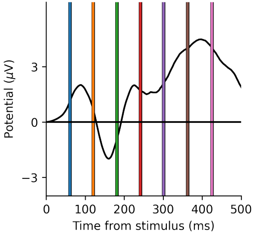

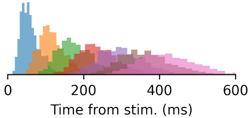

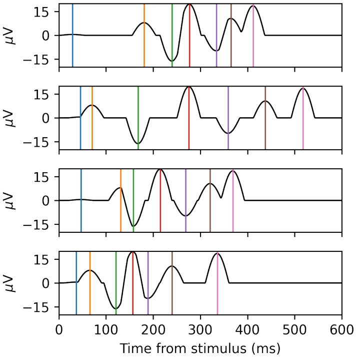

Is it only about ERPs?

Other examples of time contamination:

- Time-frequency Sherman et al., 2016
- fMRI Mumford et al., 2024

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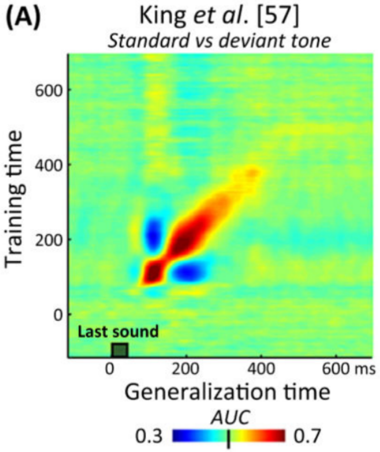


Figure: King and Dehaene, 2014

HsMM-MVPA

Hidden semi-Markov Model Multivariate Pattern Analysis (Anderson et al., 2016)

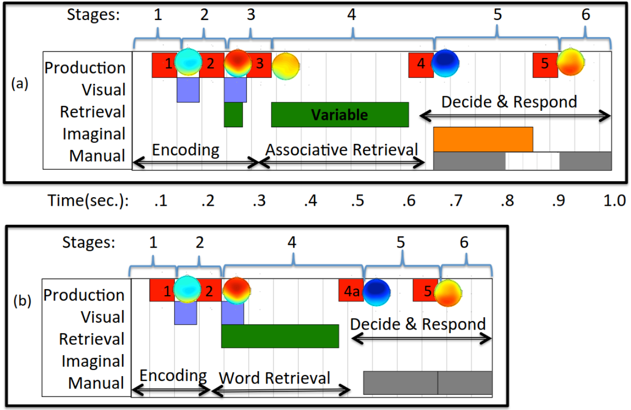
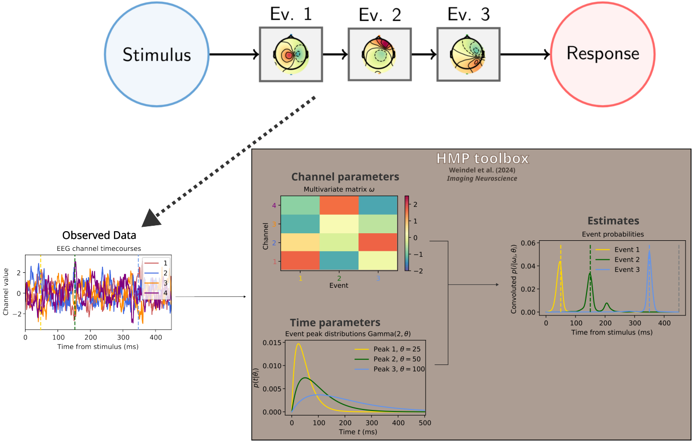


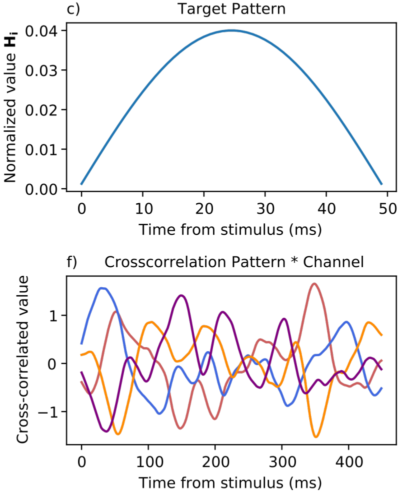
Figure: Anderson et al., 2016

Hidden multivariate pattern model

A generalization of HsMM-MVPA: **Hidden Multivariate Pattern model** (Weindel et al., 2024)



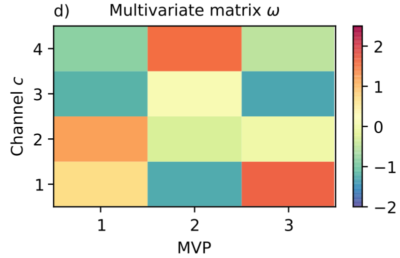
Step-by-step method: Cross-correlation

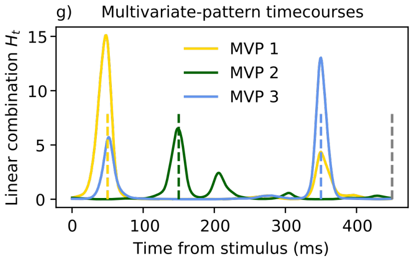


Dot product of the value of the sample s at time t for channel c with the pattern of length L:

$$h_{tc} = \sum_{l=0}^{L-1} s_{t+l,c}\mathbf{H}_l$$

Step-by-step method: MVPA

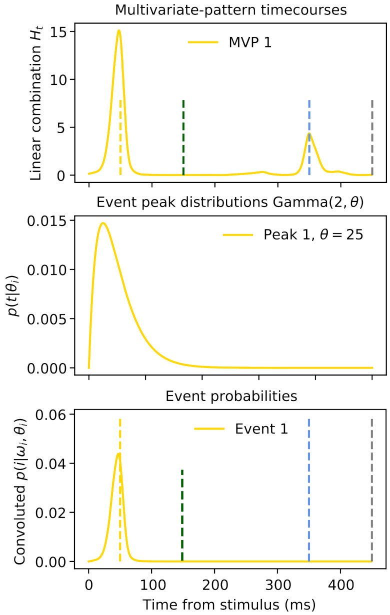




The match between the channels and the event i at time t:

$$H_{ti} = \exp\left(\sum_{c=1}^{C} h_{tc}\omega_{ci} - \frac{\omega_{ci}^2}{2}\right)$$

Step-by-step method: convolution (with one event)



When N event = 1

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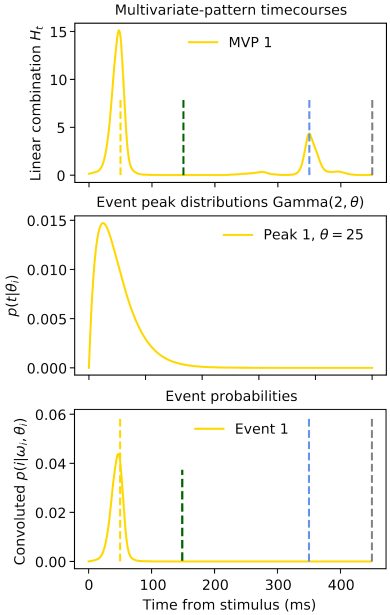


When N event = 1

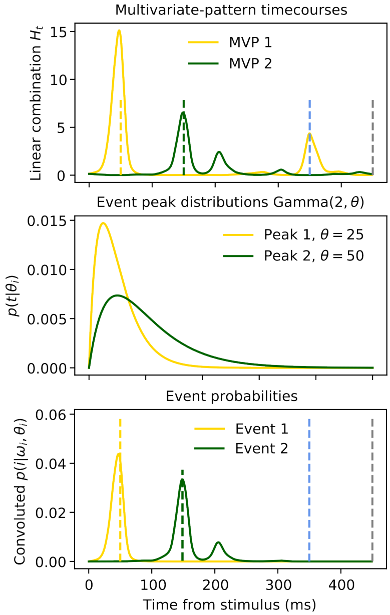
$$p(t|\theta) = \frac{g(t;\theta)}{\sum_{j=0}^{T} g(t_j;\theta)}$$

Where g is the PDF of a Gamma distribution with a shape of 2 and a scale of θ

$$p(i = t|\omega,\theta) = H_{ti}\, p(t-1|\theta)\, p(T-t|T-\theta)$$

Thus the probability that the peak of Event i is at time sample t is given by the product of the MVPA match and the time distribution of the event.

Step-by-step method: convolution(s)



When N event is > 1, account for previous event

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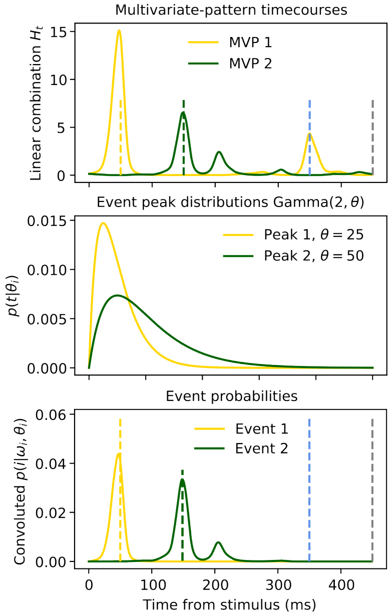

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$$\alpha_t(i) = \begin{cases} p(t|\theta_i) H_{ti} & \text{for } i = 1 \\ (\alpha_{1:t}(i-1) * p(t|\theta_i))\, H_{ti} & \text{for } 1 < i < I \end{cases}$$

$$\beta_t(i) = \begin{cases} p(t|\theta_i) & \text{for } i = I \\ (\beta_{T:t}(i+1) * p(t|\theta_i)) H_{ti} & \text{for } 0 < i < I \end{cases}$$

$$L[i = t|\theta_i, \omega_i] = \alpha_t(i)\beta_t(i)$$

$$p(i,t) = \frac{L[i = t|\theta_i, \omega_i]}{\sum_{t=1}^{T} L[i = t|\theta_i, \omega_i]}$$

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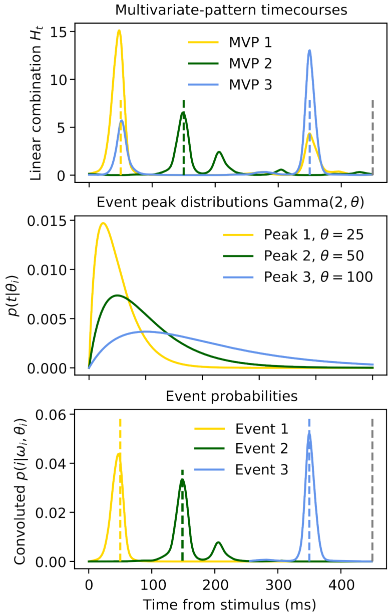

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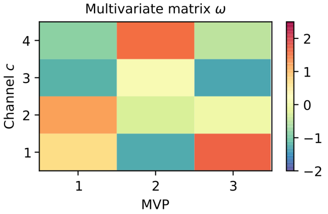

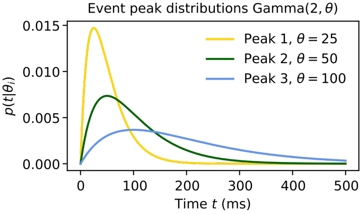

Make a proposal for the set of parameters (λ_1):
$I + 1$ peak times θ and $I \times C$ magnitudes ω

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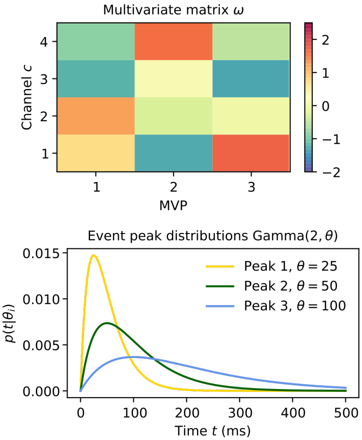

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$$LL = \log(\sum_t^T [L_t|\lambda_1])$$

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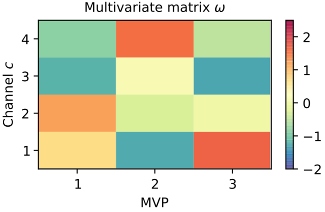

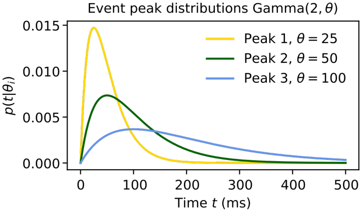

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Compute new $\lambda*$ on event probabilities:

$$\theta_i^* = \begin{cases} f(\frac{\sum_1^T p(i,t)t}{T}) & \text{for } i = 1 \\ f(\frac{\sum_1^T p(i,t)t}{T} - \frac{\sum_1^T p(i-1,t)t}{T}) & \text{for } 1 < i \leq I \end{cases}$$

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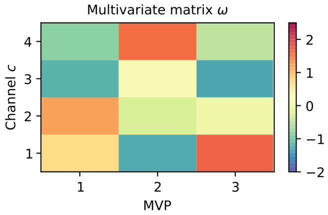

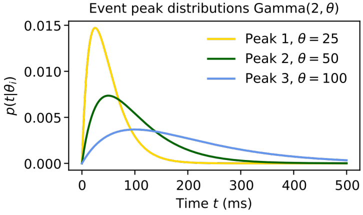

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End when $\epsilon > (LL_{n+1} - LL_n)/|LL_n|$

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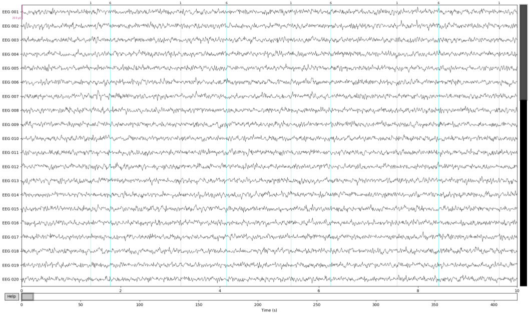
- Physiological time-series (EEG, MEG, EcOG, iEEG?, ...)
- Parsing sequence start and end (e.g., stim to response)
- PCA + cross-correlation with expected pattern



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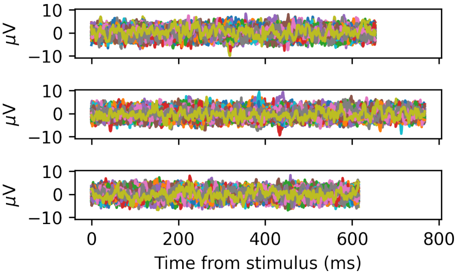

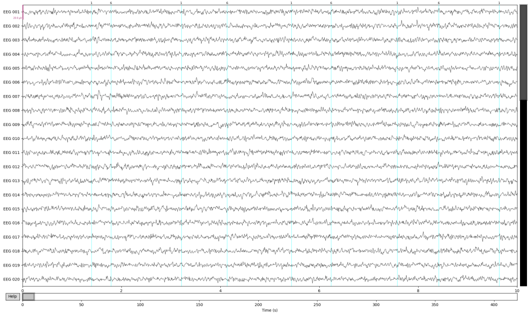

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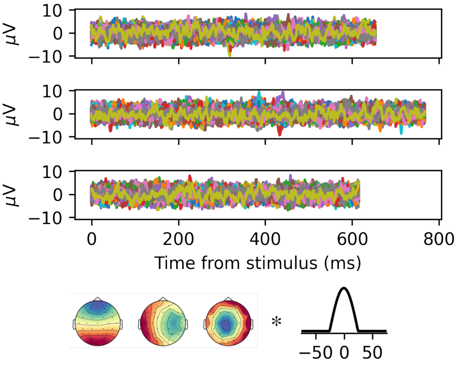

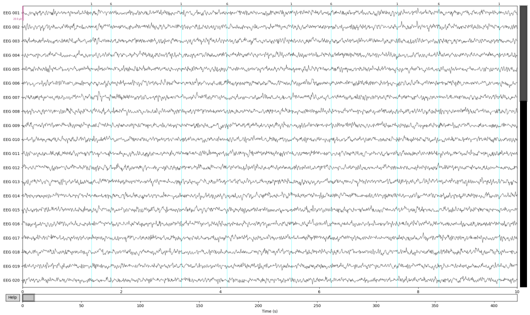

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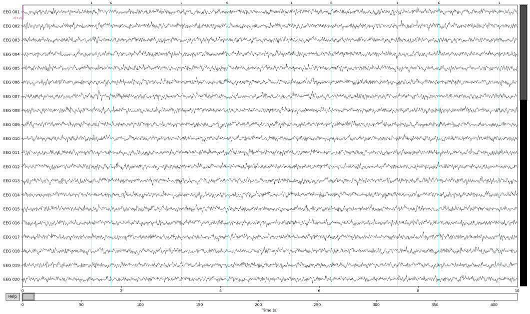

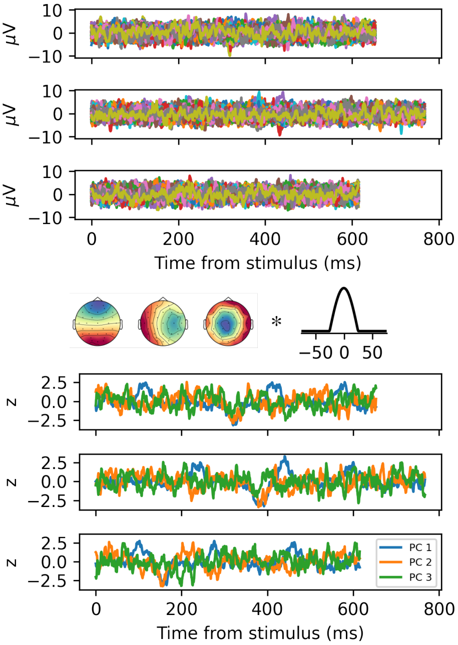

How many events?

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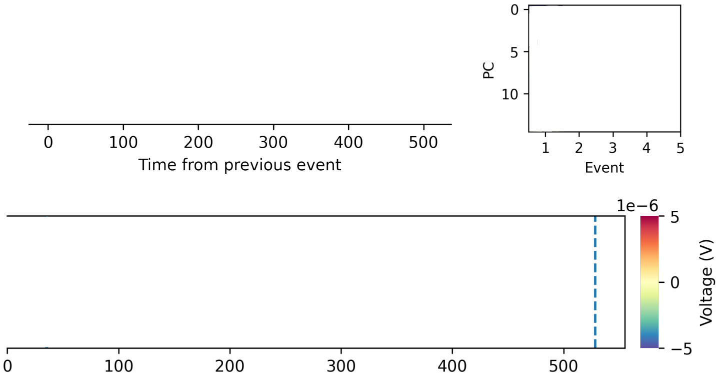

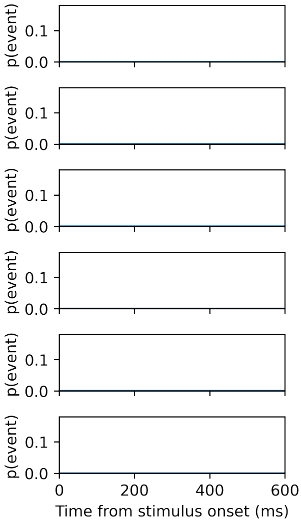

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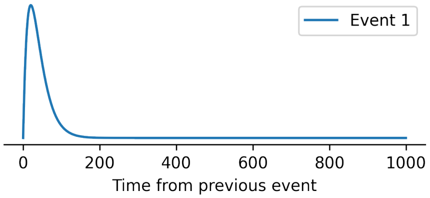

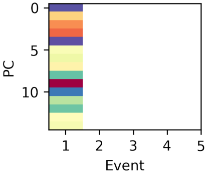

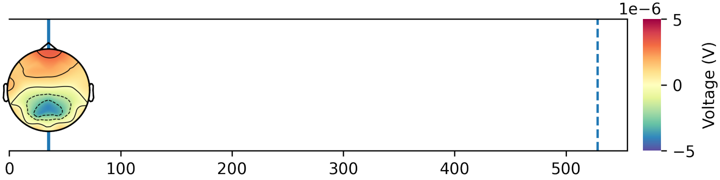

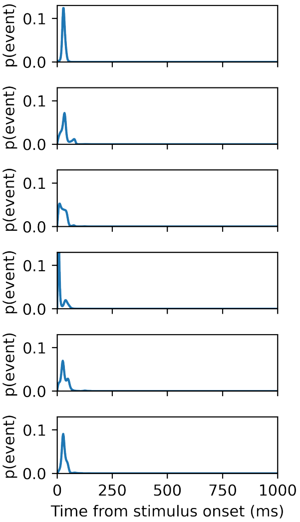

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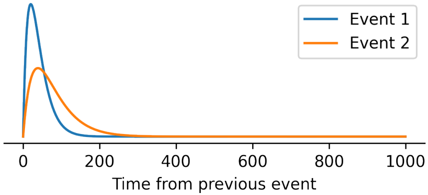

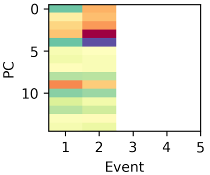

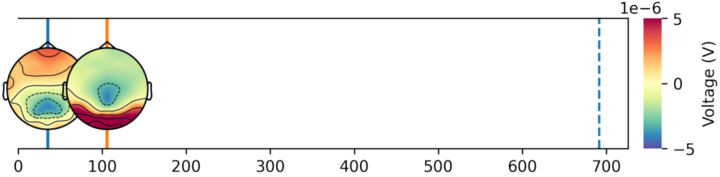

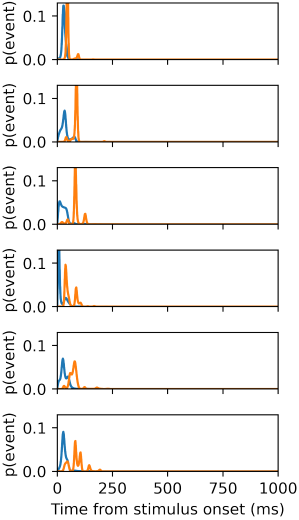

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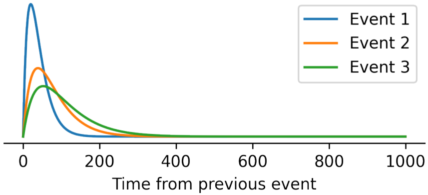

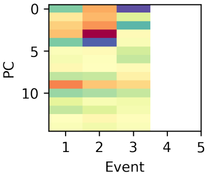

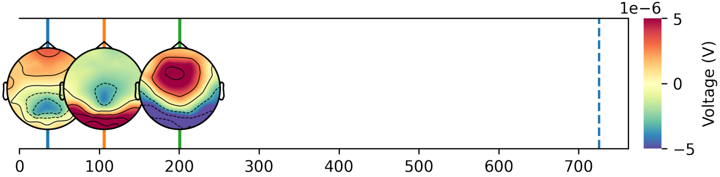

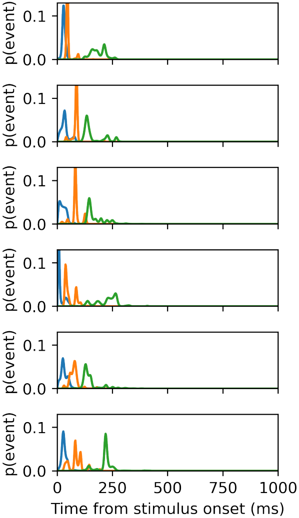

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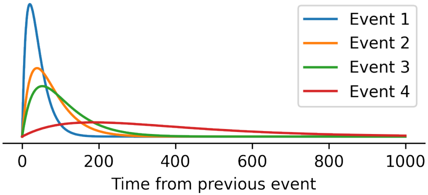

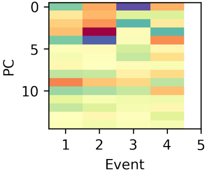

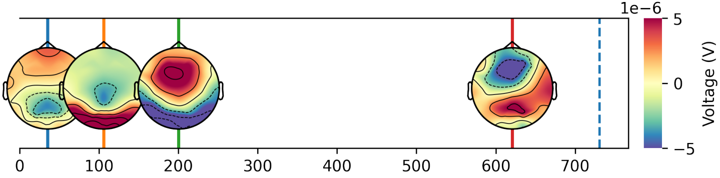

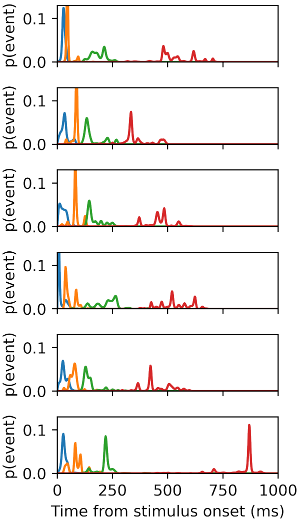

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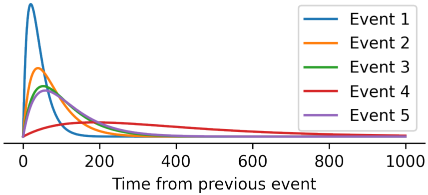

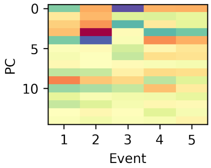

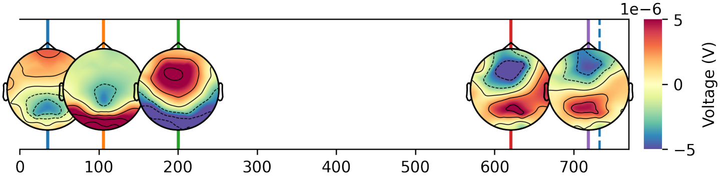

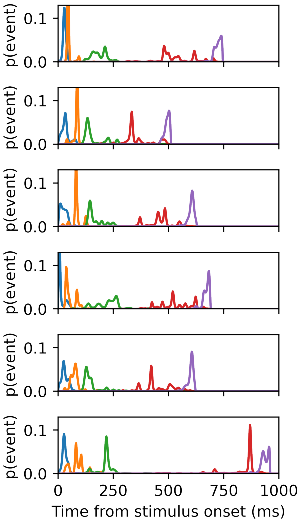

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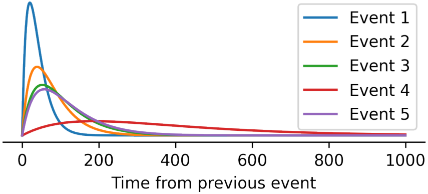
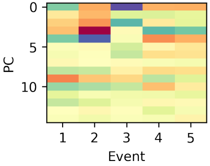
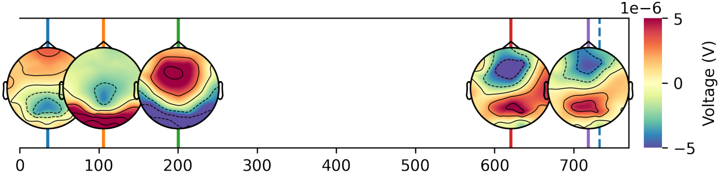
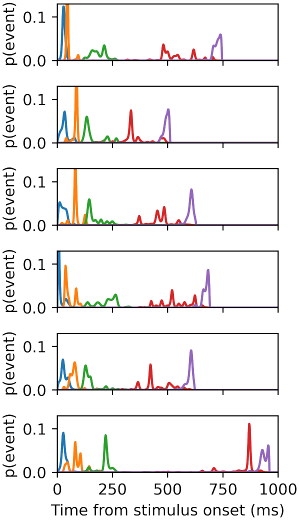




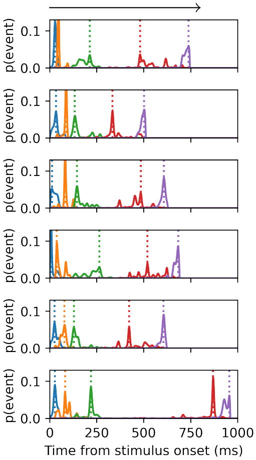




Alternatively: fit an I event model. With appropriate signal-to-noise ratio, the solution with I inferred events should be the same as directly estimating I events.

Estimating single-trial times: inspecting time

argmax(p(Ev.,trial))



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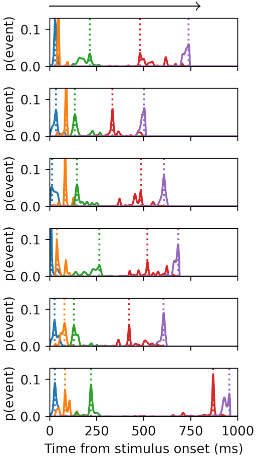

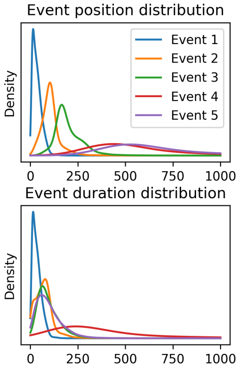

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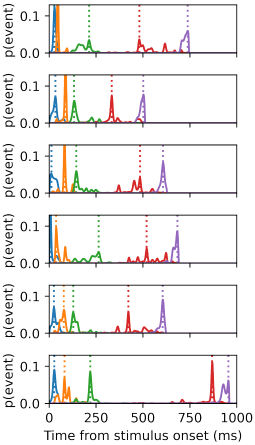
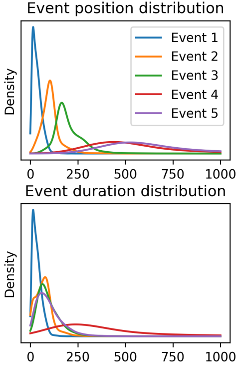
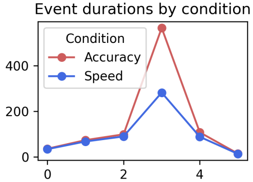
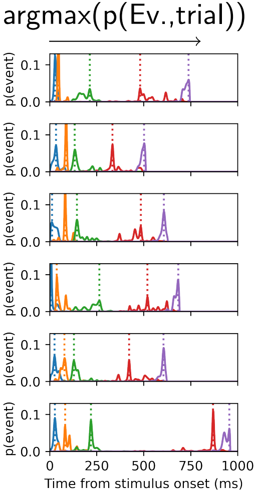
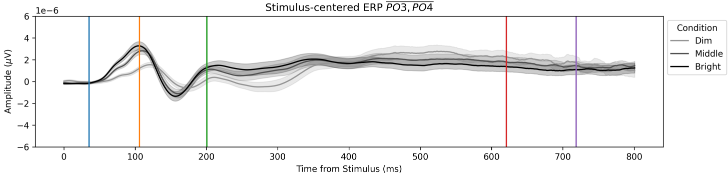






Estimating single-trial times: inspecting electrophysiology





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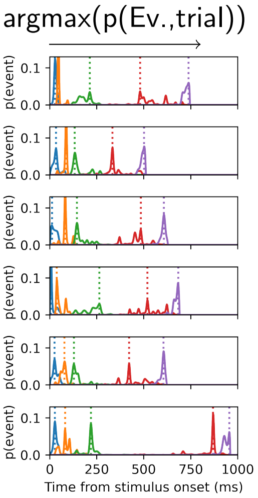
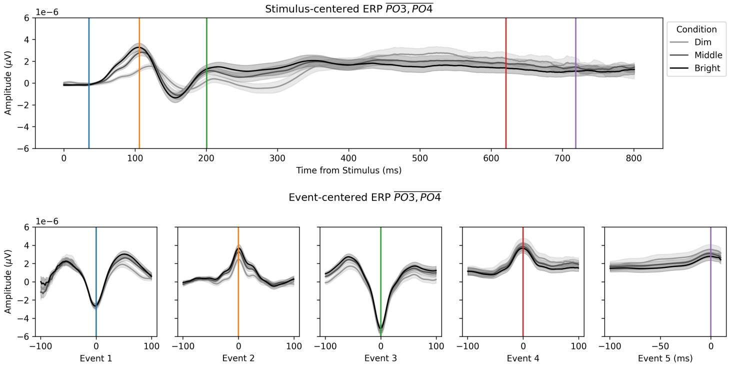
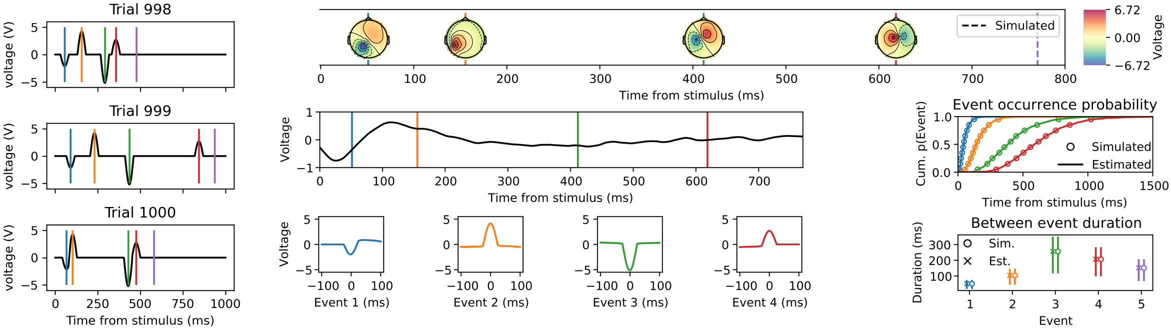




Example in simulated data

Back to noiseless simulated data:



Core assumptions

Three core assumptions

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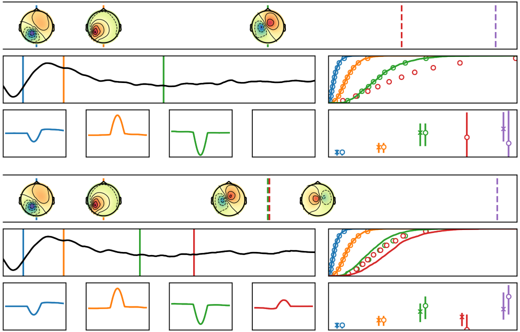

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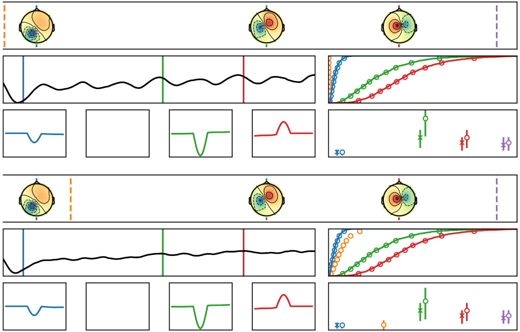

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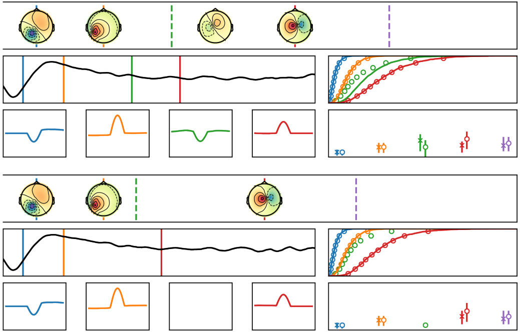

Core assumptions

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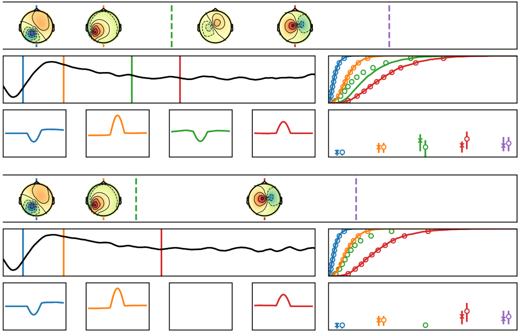
- Events in the interval are sequential
- Events are transient
- Events are repeated across trials
- (The modelled durations contain the sequence)



A note on ancillary assumptions

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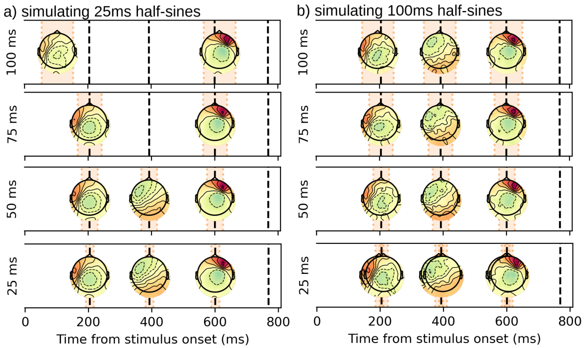
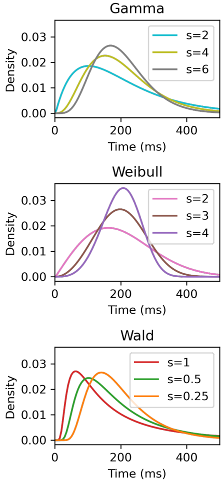


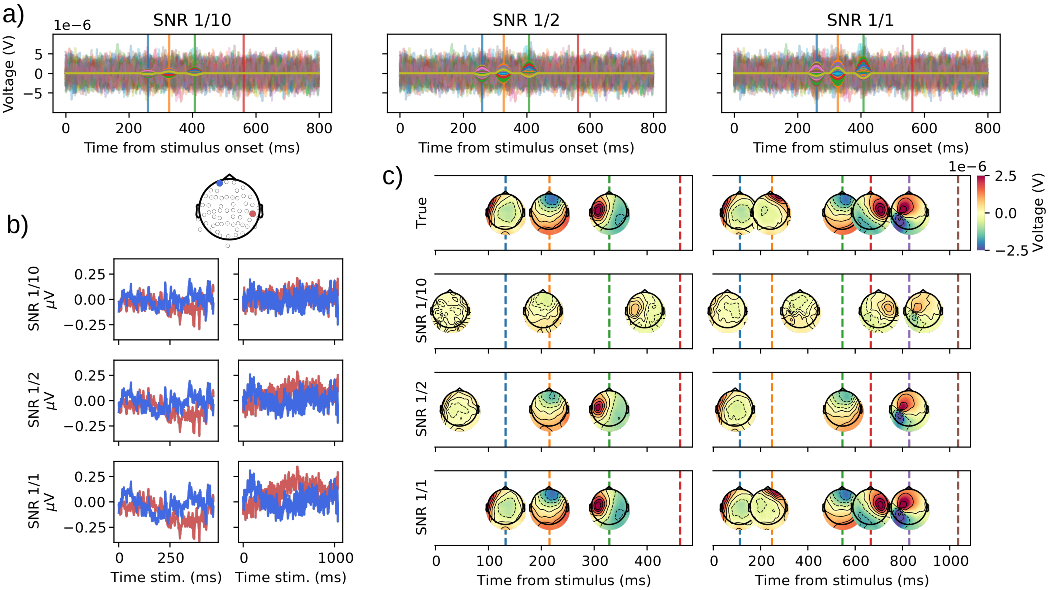
A note on ancillary assumptions

Ancillary assumptions:

- Events are equal or shorter than the expected duration
- The time distribution family doesn't exclude likely values



A note on signal-to-noise ratio



HMP and the P3

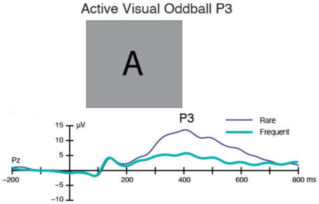


Figure: Adapted from ERP Core (Kappenman et al., 2021)

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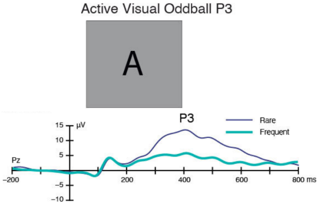
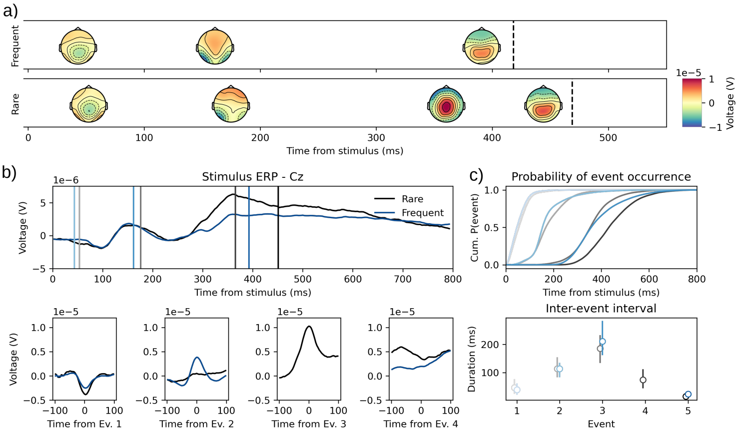
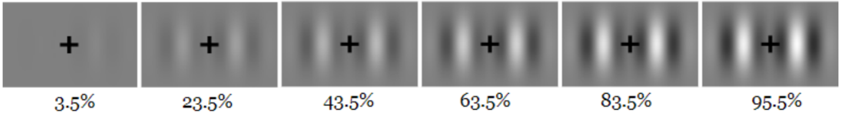


Figure: Adapted from ERP Core (Kappenman et al., 2021)



Application to decision-making: time-domain

Application to a decision making task (Weindel et al., 2025)



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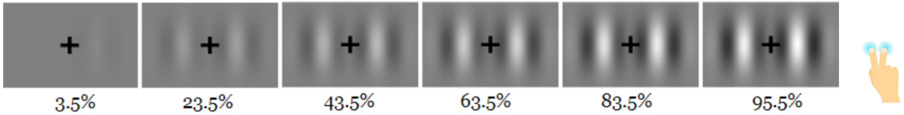

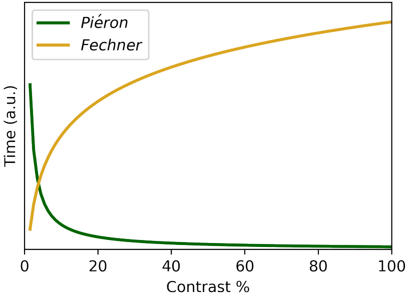

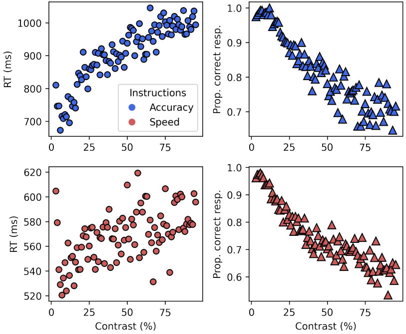

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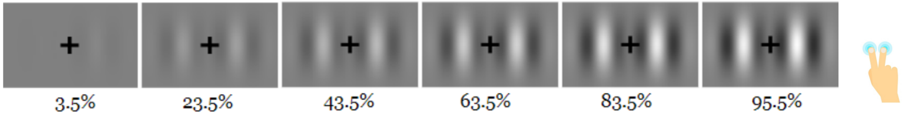
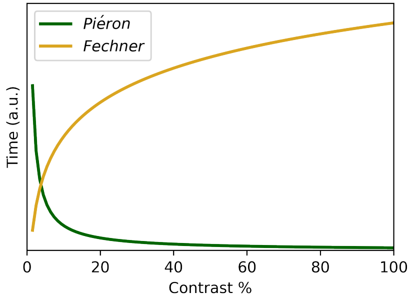
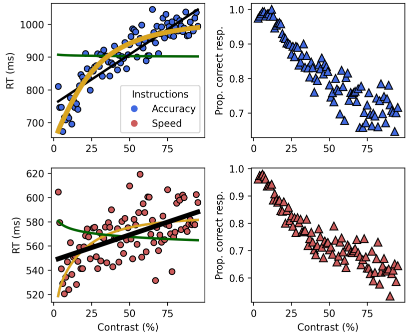
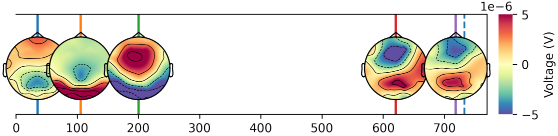






EEG results: inter-event timing



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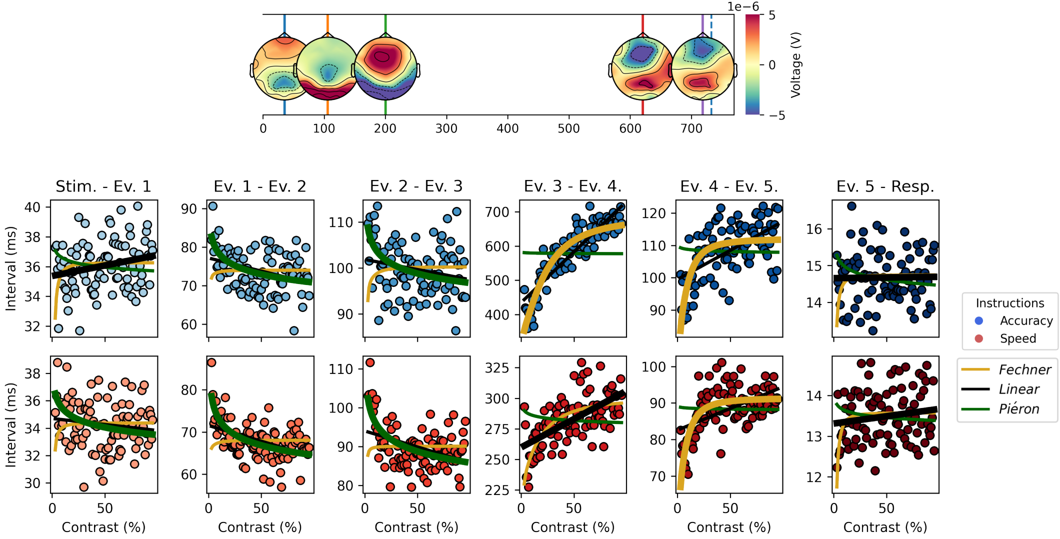
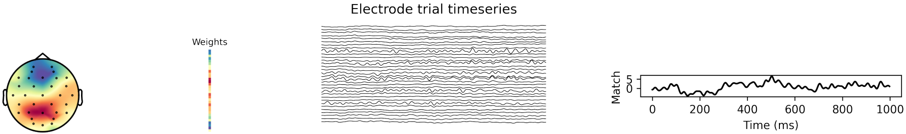
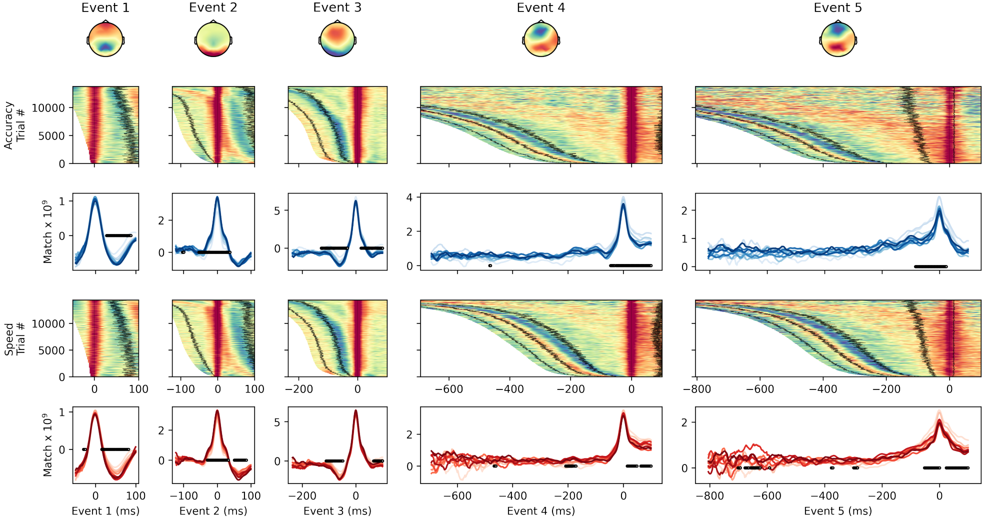


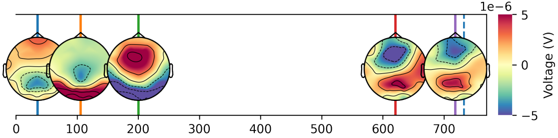
EEG results: Electrophysiological activity I



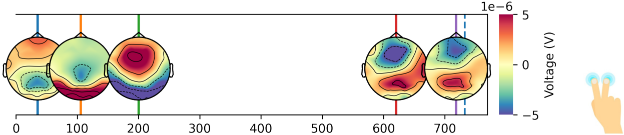
EEG results: Electrophysiological activity II

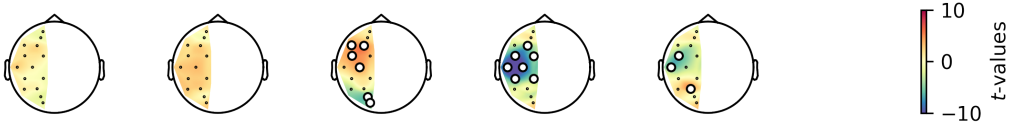


EEG results: Electrophysiological activity III

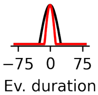


EEG results: Electrophysiological activity III

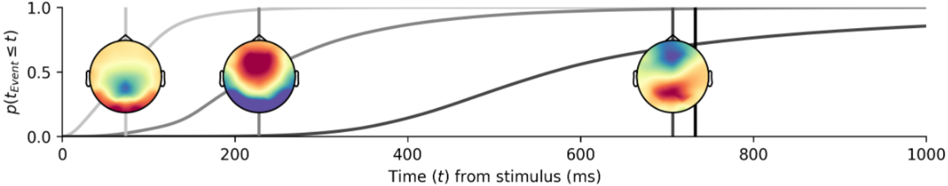




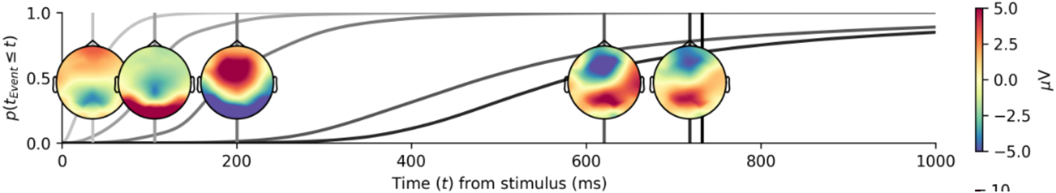
A note on pattern width



https://elifesciences.org/reviewed-preprints/108049v1:



https://www.biorxiv.org/content/10.1101/2025.06.25.661505v2:



Perks

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IFF all assumptions are correct, we access the single-trial time of information-processing related components in EEG and thus we can:

- Decompose any duration (saccade, key strokes, movement time) into task-relevant sub-intervals
- Describe tasks and conditions based on an underlying sequence of components
- Describe components in time and space
- Time-lock follow-up analyses on specific events (source reconstruction, time-frequency,...)
- Use task-derived spatial filters

Advices (yet unwritten)

Note that the model and its implementation are work in progress. At this stage, here are a few recommendations based on user experiences:

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Note that the model and its implementation are work in progress. At this stage, here are a few recommendations based on user experiences:

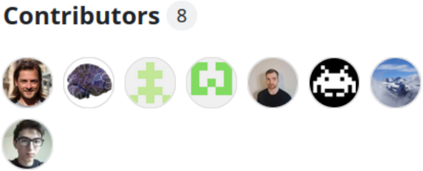
- Use design with strong hypotheses to constrain the interpretation (no HARKing).
- Start with relatively short intervals (e.g. avoid long 5 seconds durations)
- Start exploring with low precision on downsampled data (not less than 100Hz, 250Hz recommended)
- If inferring events from data, keep your amount of trials as high as possible (e.g. group model > participant model)
- Don't waste too much information at the preprocessing step (bad data rejection, ICA, PCA)

Ressources

A list of resources:

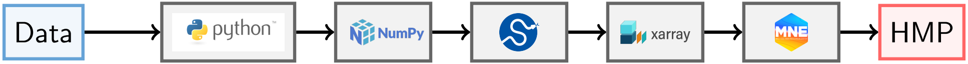
- **Tutorials** https://hmp.readthedocs.io/en/latest/notebooks/demo.html
- **Codebook/API**: https://hmp.readthedocs.io/en/latest/api/hmp.html
- **Paper on HMP**: https://direct.mit.edu/imag/article/doi/10.1162/imag_a_00400/125469
- **Repository of the decision-making study**: https://github.com/GWeindel/decision-times
- **Upcoming mailing list** (send an email)

Thank you



and you?

Thank you for being FOSS:



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